

```
In [1]: import os  
os.getcwd()
```

```
Out[1]: '/Users/anwar/Desktop/Data Camp'
```

```
In [2]: import pandas as pd  
  
df = pd.read_csv("/Users/anwar/Desktop/Data Camp/retail_black_friday_sale  
df.head()
```

```
Out[2]:
```

	transaction_id	customer_id	age_group	gender	city	customer_segment
0	T0000001	C007297	26-35	Male	San Francisco	Loyal
1	T0000002	C001640	56+	Other	Dallas	Returning
2	T0000003	C018025	26-35	Male	Phoenix	New
3	T0000004	C016050	18-25	Other	Miami	VIP
4	T0000005	C014629	46-55	Male	Los Angeles	Loyal

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   transaction_id         100000 non-null  object
1   customer_id            100000 non-null  object
2   age_group              100000 non-null  object
3   gender                 100000 non-null  object
4   city                   100000 non-null  object
5   customer_segment       100000 non-null  object
6   product_id             100000 non-null  object
7   product_category       100000 non-null  object
8   original_price          100000 non-null  float64
9   discount_pct           100000 non-null  int64
10  final_price            100000 non-null  float64
11  quantity               100000 non-null  int64
12  purchase_amount         100000 non-null  float64
13  payment_method          100000 non-null  object
14  purchase_date           100000 non-null  object
15  purchase_hour           100000 non-null  int64
16  is_weekend              100000 non-null  int64
17  is_black_friday         100000 non-null  int64
dtypes: float64(3), int64(5), object(10)
memory usage: 13.7+ MB
```

In [5]: `df.shape`

Out[5]: (100000, 18)

In [6]: `df.describe()`

Out[6]:

	original_price	discount_pct	final_price	quantity	purchase_a
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.0
mean	330.281911	26.631150	242.220661	1.447290	351.2
std	444.315916	12.249229	332.677331	0.815762	589.8
min	5.000000	5.000000	2.070000	1.000000	2.2
25%	88.117500	20.000000	62.940000	1.000000	75.7
50%	180.220000	25.000000	129.490000	1.000000	161.1
75%	333.712500	35.000000	247.575000	2.000000	358.7
max	2499.890000	60.000000	2374.900000	5.000000	11441.8

In [7]: `df.isnull().sum()`

```
Out[7]: transaction_id      0
        customer_id       0
        age_group         0
        gender            0
        city              0
        customer_segment  0
        product_id        0
        product_category   0
        original_price     0
        discount_pct       0
        final_price        0
        quantity          0
        purchase_amount    0
        payment_method     0
        purchase_date      0
        purchase_hour      0
        is_weekend         0
        is_black_friday    0
        dtype: int64
```

```
In [8]: df["gender"].value_counts()
```

```
Out[8]: gender
        Female    33460
        Male      33396
        Other     33144
        Name: count, dtype: int64
```

```
In [9]: df["city"].value_counts()
```

```
Out[9]: city
        New York    10105
        Seattle     10097
        Atlanta     10075
        Miami       10036
        Phoenix     10032
        Houston     10024
        Dallas      9990
        San Francisco 9968
        Chicago     9954
        Los Angeles 9719
        Name: count, dtype: int64
```

```
In [10]: df["product_category"].value_counts()
```

```
Out[10]: product_category
Accessories      10156
Footwear         10104
Toys             10054
Books            10031
Beauty           9984
Sports           9975
Electronics      9974
Groceries        9925
Clothing         9911
Home & Kitchen   9886
Name: count, dtype: int64
```

```
In [11]: df["age_group"].value_counts()
```

```
Out[11]: age_group
36-45      20145
46-55      20014
18-25      19974
26-35      19960
56+        19907
Name: count, dtype: int64
```

```
In [12]: df["payment_method"].value_counts()
```

```
Out[12]: payment_method
Debit Card    16772
Mobile Wallet 16711
Cash          16680
Credit Card   16663
Gift Card     16614
PayPal        16560
Name: count, dtype: int64
```

```
In [13]: df.duplicated().sum()
```

```
Out[13]: np.int64(0)
```

```
In [14]: df.dtypes
```

```
Out[14]: transaction_id      object
customer_id      object
age_group        object
gender           object
city             object
customer_segment object
product_id       object
product_category object
original_price    float64
discount_pct     int64
final_price      float64
quantity         int64
purchase_amount  float64
payment_method   object
purchase_date    object
purchase_hour    int64
is_weekend       int64
is_black_friday  int64
dtype: object
```

```
In [15]: df["gender"].unique()
```

```
Out[15]: array(['Male', 'Other', 'Female'], dtype=object)
```

```
In [16]: df["city"].unique()
```

```
Out[16]: array(['San Francisco', 'Dallas', 'Phoenix', 'Miami', 'Los Angeles',
               'Seattle', 'New York', 'Chicago', 'Houston', 'Atlanta'],
              dtype=object)
```

```
In [17]: df["customer_segment"].unique()
```

```
Out[17]: array(['Loyal', 'Returning', 'New', 'VIP'], dtype=object)
```

```
In [18]: df["product_category"].unique()
```

```
Out[18]: array(['Footwear', 'Beauty', 'Groceries', 'Sports', 'Accessories',
               'Home & Kitchen', 'Toys', 'Clothing', 'Electronics', 'Books'],
              dtype=object)
```

```
In [22]: for col in ["gender",
                    "age_group",
                    "product_category",
                    "payment_method",
                    "city",
                    "customer_segment"]:
    print("\n" + col)
    print(df[col].value_counts())
```

```
gender
gender
Female    33460
Male      33396
```

Other 33144
Name: count, dtype: int64

age_group
age_group
36-45 20145
46-55 20014
18-25 19974
26-35 19960
56+ 19907
Name: count, dtype: int64

product_category
product_category
Accessories 10156
Footwear 10104
Toys 10054
Books 10031
Beauty 9984
Sports 9975
Electronics 9974
Groceries 9925
Clothing 9911
Home & Kitchen 9886
Name: count, dtype: int64

payment_method
payment_method
Debit Card 16772
Mobile Wallet 16711
Cash 16680
Credit Card 16663
Gift Card 16614
PayPal 16560
Name: count, dtype: int64

city
city
New York 10105
Seattle 10097
Atlanta 10075
Miami 10036
Phoenix 10032
Houston 10024
Dallas 9990
San Francisco 9968
Chicago 9954
Los Angeles 9719
Name: count, dtype: int64

customer_segment
customer_segment
Returning 34946

```
New          29956
Loyal        25130
VIP           9968
Name: count, dtype: int64
```

```
In [25]: df.groupby("gender")["purchase_amount"].sum().sort_values(ascending=False)
```

```
Out[25]: gender
Male      11767285.96
Other     11719413.02
Female    11638931.69
Name: purchase_amount, dtype: float64
```

```
In [26]: df.groupby("city")["purchase_amount"].sum().sort_values(ascending=False)
```

```
Out[26]: city
Atlanta      3618348.52
Miami        3566829.80
Dallas       3547967.43
New York     3546500.70
Phoenix      3526014.43
Chicago      3512360.63
Houston      3474957.45
Seattle      3456842.39
San Francisco 3445376.30
Los Angeles  3430433.02
Name: purchase_amount, dtype: float64
```

```
In [27]: df.groupby("product_category")["purchase_amount"].sum().sort_values(ascen
```

```
Out[27]: product_category
Electronics  14123441.47
Home & Kitchen  6461099.92
Sports       3303563.29
Accessories  2754561.54
Footwear     2013317.39
Clothing     1688471.88
Toys         1669209.91
Beauty       1374751.97
Groceries    1075986.12
Books        661227.18
Name: purchase_amount, dtype: float64
```

```
In [28]: df.groupby("payment_method")["purchase_amount"].sum().sort_values(ascendi
```

```
Out[28]: payment_method
Mobile Wallet  6020177.60
Debit Card    5892551.34
Credit Card   5877297.96
Cash          5856797.42
PayPal        5770589.41
Gift Card     5708216.94
Name: purchase_amount, dtype: float64
```

```
In [29]: df["purchase_amount"].sum()
```

```
Out[29]: np.float64(35125630.67)
```

```
In [31]: df["purchase_amount"].mean().round(2)
```

```
Out[31]: np.float64(351.26)
```

```
In [33]: df.groupby("product_category")["quantity"].sum().sort_values(ascending=False)
```

```
Out[33]: product_category
Accessories      14732
Toys             14663
Footwear        14529
Sports          14503
Electronics     14502
Books           14437
Home & Kitchen  14393
Beauty          14385
Clothing        14317
Groceries       14268
Name: quantity, dtype: int64
```

```
In [35]: df["discount_pct"].mean().round(2)
```

```
Out[35]: np.float64(26.63)
```

```
In [39]: df["discount_amount"] = df["original_price"] - df["final_price"]
```

```
In [48]: df.head()
```

```
Out[48]:
```

	transaction_id	customer_id	age_group	gender	city	customer_segment
0	T0000001	C007297	26-35	Male	San Francisco	Loyal
1	T0000002	C001640	56+	N/A	Dallas	Returning
2	T0000003	C018025	26-35	Male	Phoenix	New
3	T0000004	C016050	18-25	N/A	Miami	VIP
4	T0000005	C014629	46-55	Male	Los Angeles	Loyal

```
In [41]: df["purchase_date"] = pd.to_datetime(df["purchase_date"])
```

```
In [42]: df["purchase_date"].dtype
```

```
Out[42]: dtype('<M8[ns]')
```



```
In [46]: df["gender"] = df["gender"].replace("other", "N/A")
```

```
In [47]: df["gender"].value_counts()
```

```
Out[47]: gender
Female    33460
Male      33396
N/A       33144
Name: count, dtype: int64
```

```
In [50]: df.groupby("customer_segment")["purchase_amount"].sum().sort_values(ascen
```

```
Out[50]: customer_segment
Returning    12125507.74
New          10574443.42
Loyal        8888416.70
VIP           3537262.81
Name: purchase_amount, dtype: float64
```

```
In [51]: df.groupby(df["purchase_date"].dt.month)["purchase_amount"].sum()
```

```
Out[51]: purchase_date
11     30712215.25
12      4413415.42
Name: purchase_amount, dtype: float64
```

```
In [52]: df.groupby("is_weekend")["purchase_amount"].sum()
```

```
Out[52]: is_weekend
0      26325138.34
1       8800492.33
Name: purchase_amount, dtype: float64
```

```
In [53]: df.groupby("is_black_friday")["purchase_amount"].sum()
```

```
Out[53]: is_black_friday
0      30743101.73
1       4382528.94
Name: purchase_amount, dtype: float64
```

```
In [55]: import matplotlib.pyplot as plt
```

```
In [57]: df["payment_method"].value_counts().sort_values(ascending=False)
```

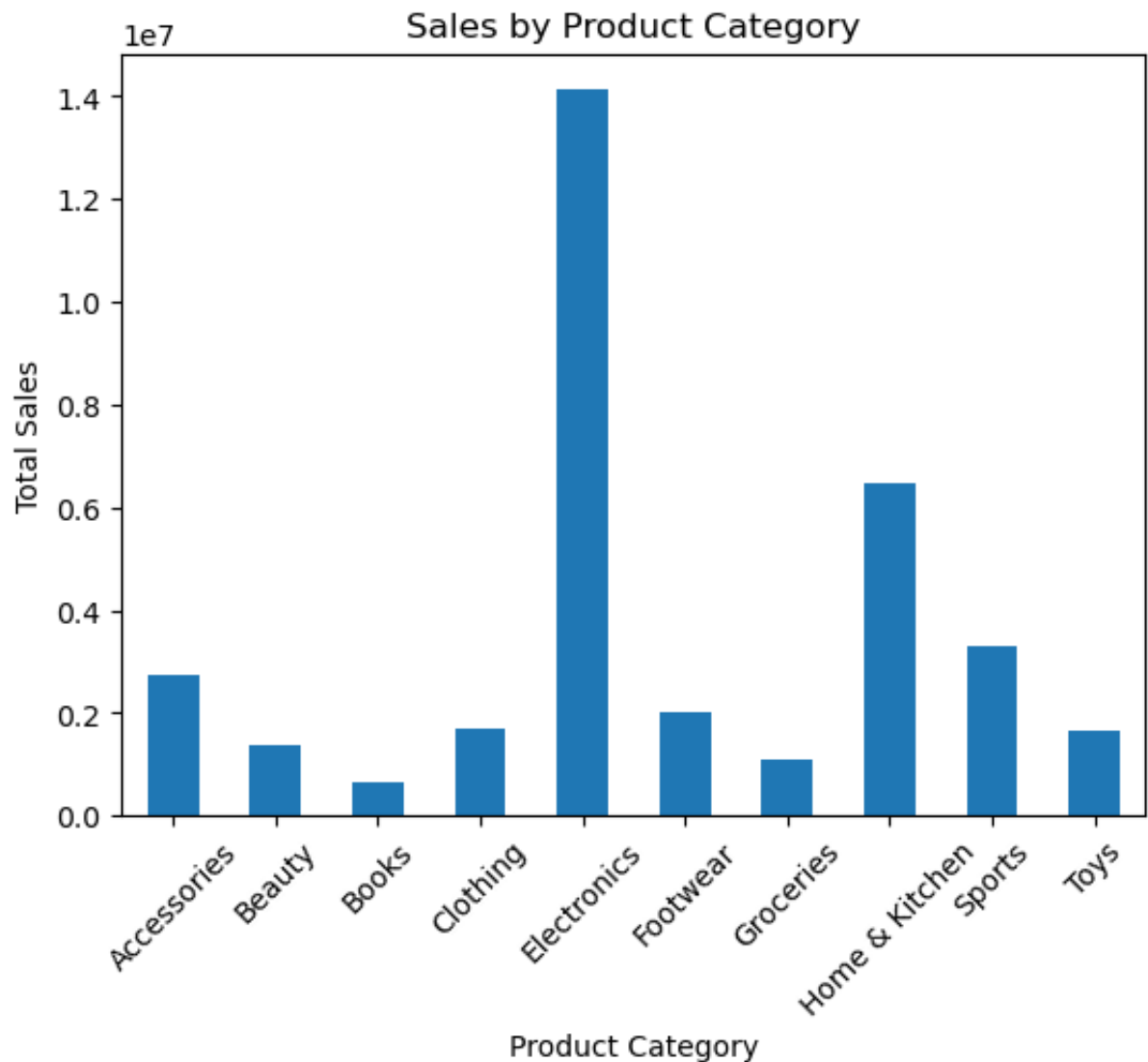
```
Out[57]: payment_method
Debit Card    16772
Mobile Wallet 16711
Cash          16680
Credit Card   16663
Gift Card     16614
PayPal        16560
Name: count, dtype: int64
```

```
In [58]: sales_by_category = df.groupby(
            "product_category"
        )["purchase_amount"].sum()

sales_by_category.plot(kind="bar")

plt.title("Sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)

plt.show()
```



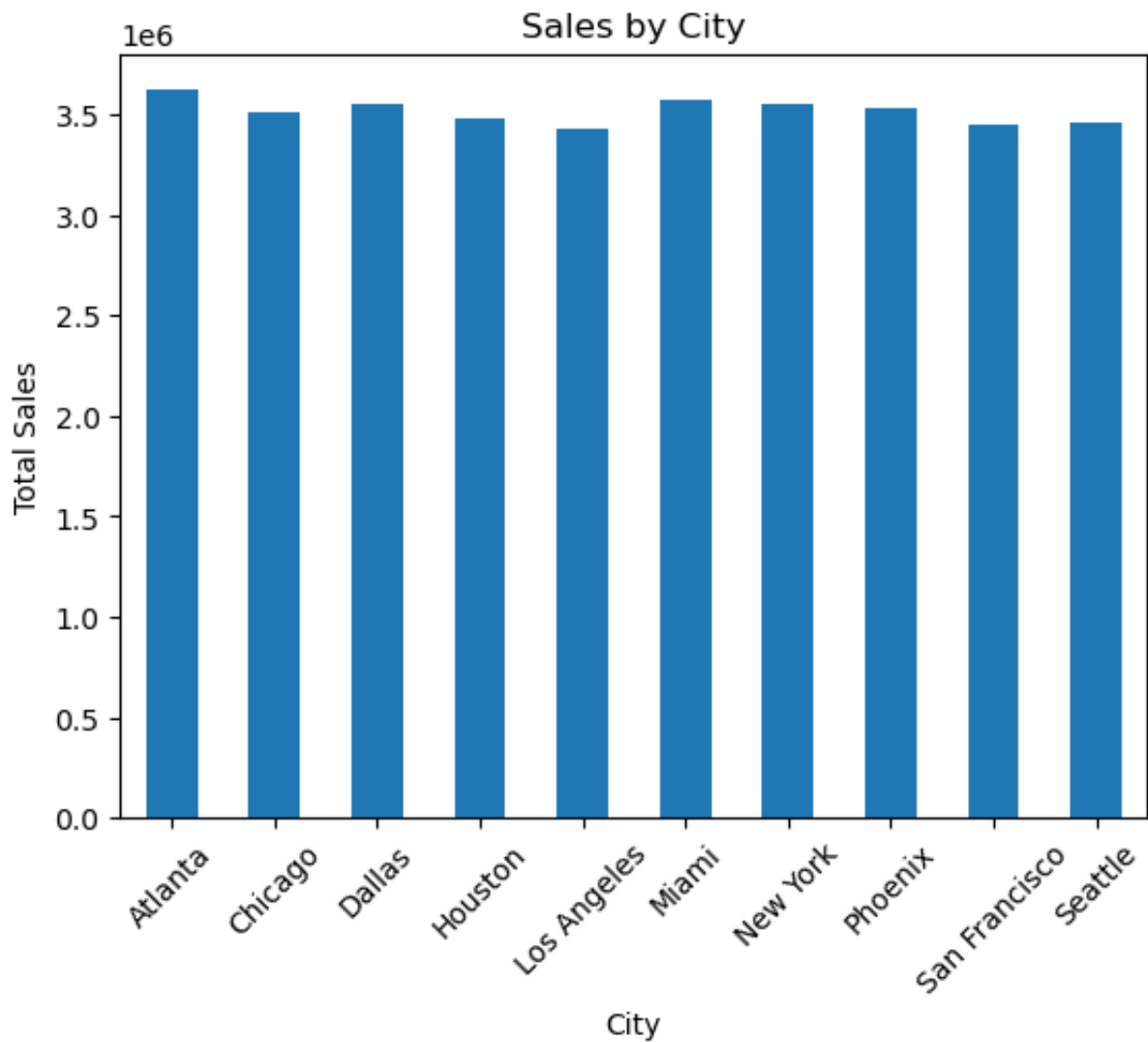
```
In [59]: sales_by_city = df.groupby(
            "city"
        )["purchase_amount"].sum()

sales_by_city.plot(kind="bar")

plt.title("Sales by City")
plt.xlabel("City")
```

```
plt.ylabel("Total Sales")
plt.xticks(rotation=45)

plt.show()
```

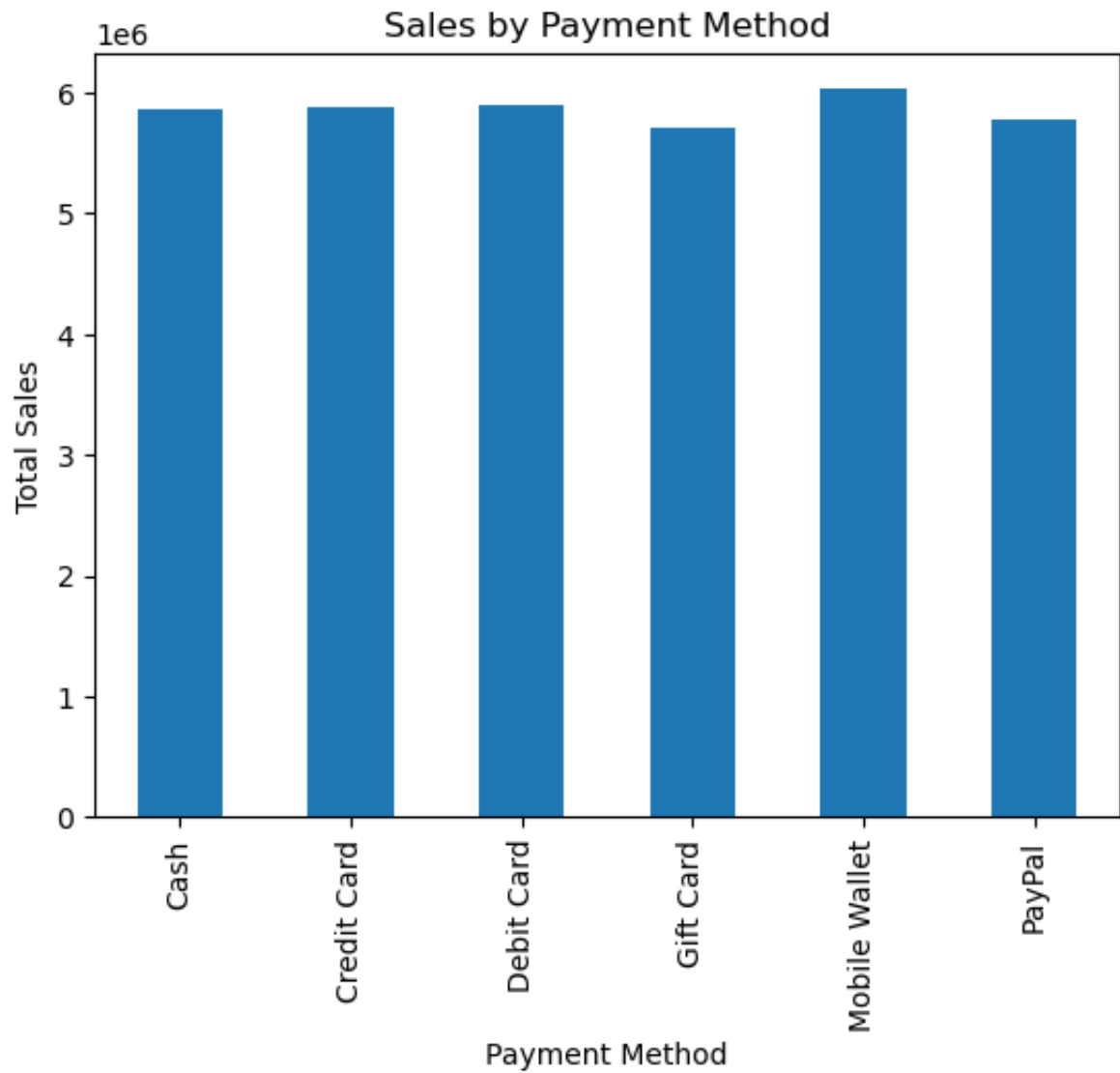


```
In [60]: sales_by_payment = df.groupby(
        "payment_method"
    )["purchase_amount"].sum()

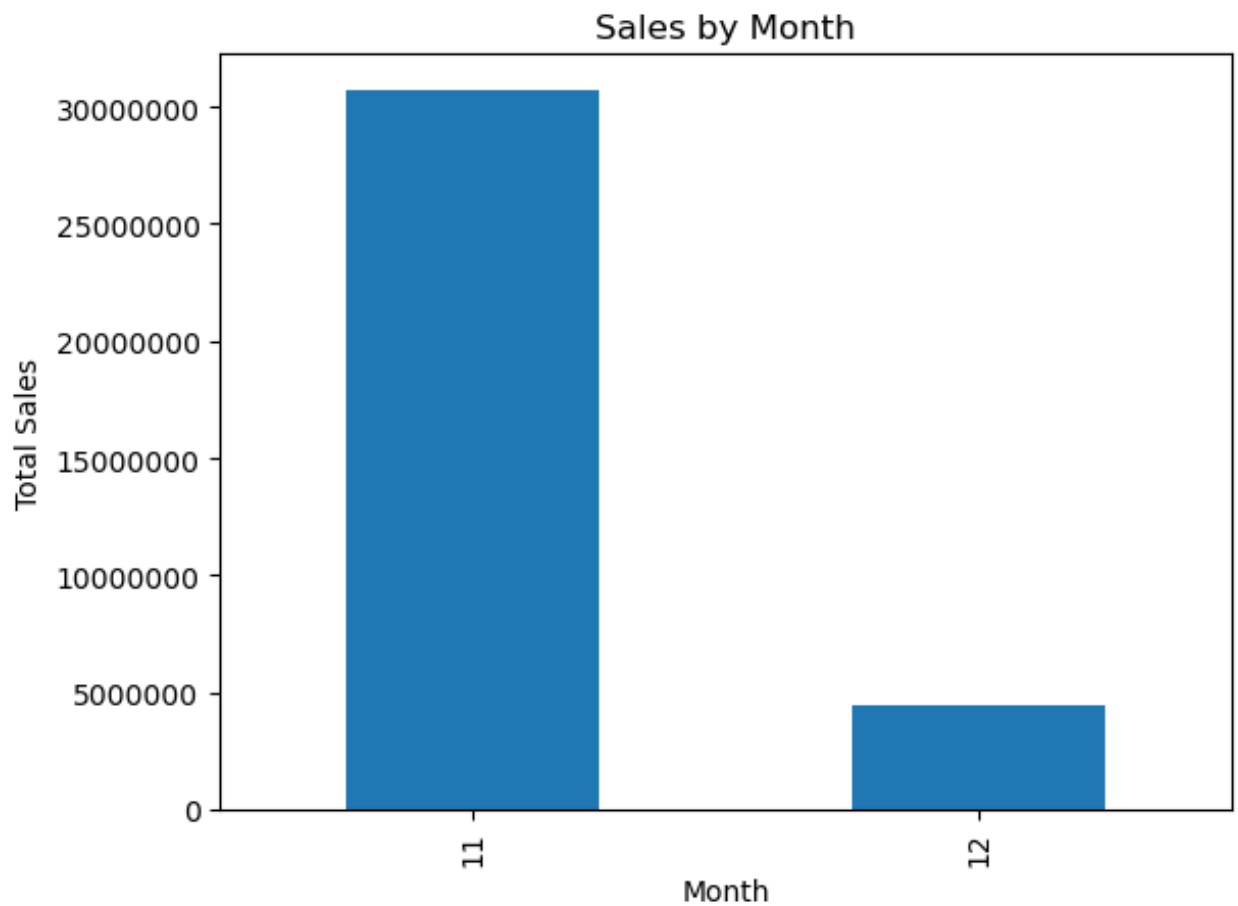
sales_by_payment.plot(kind="bar")

plt.title("Sales by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Total Sales")

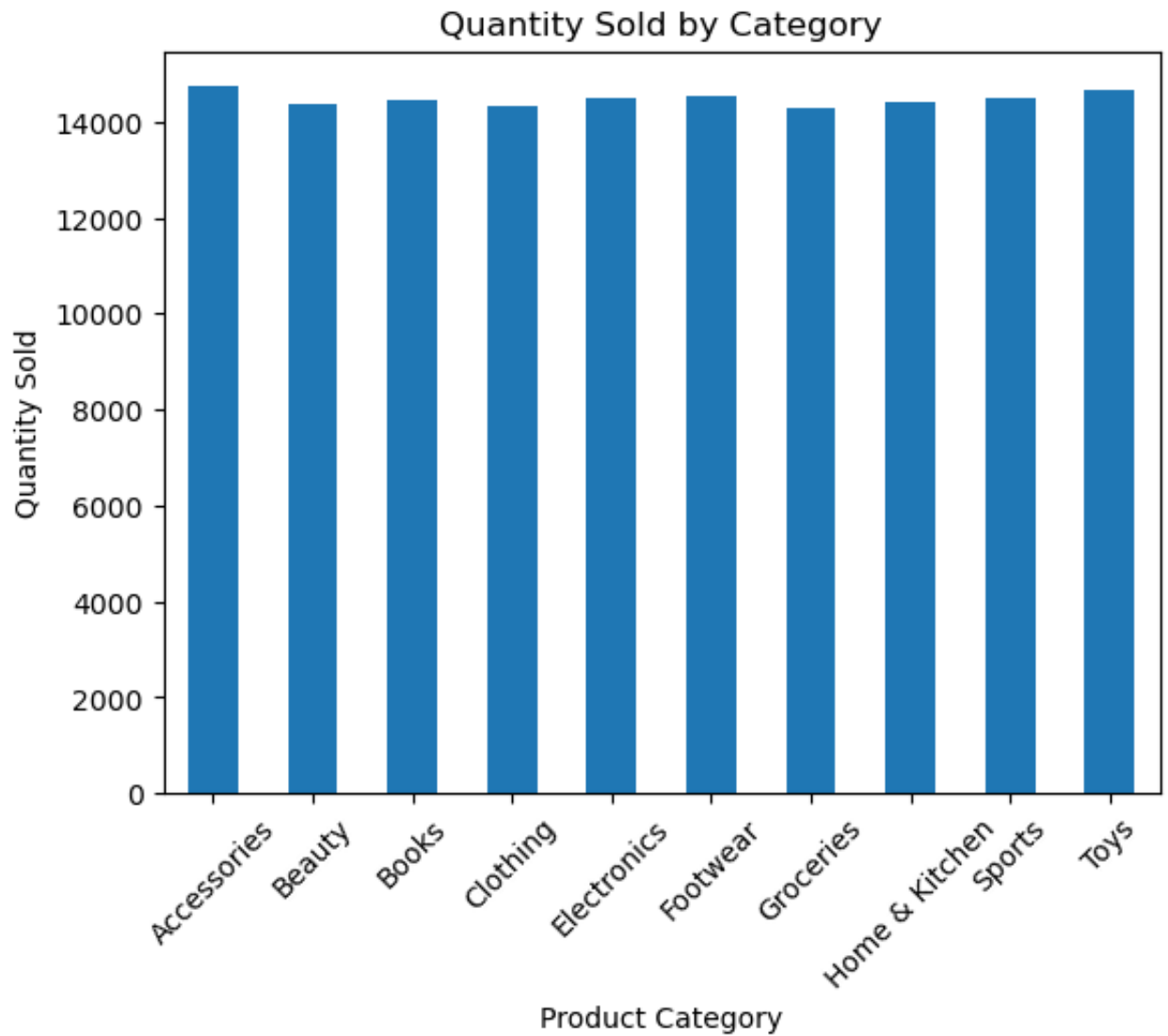
plt.show()
```



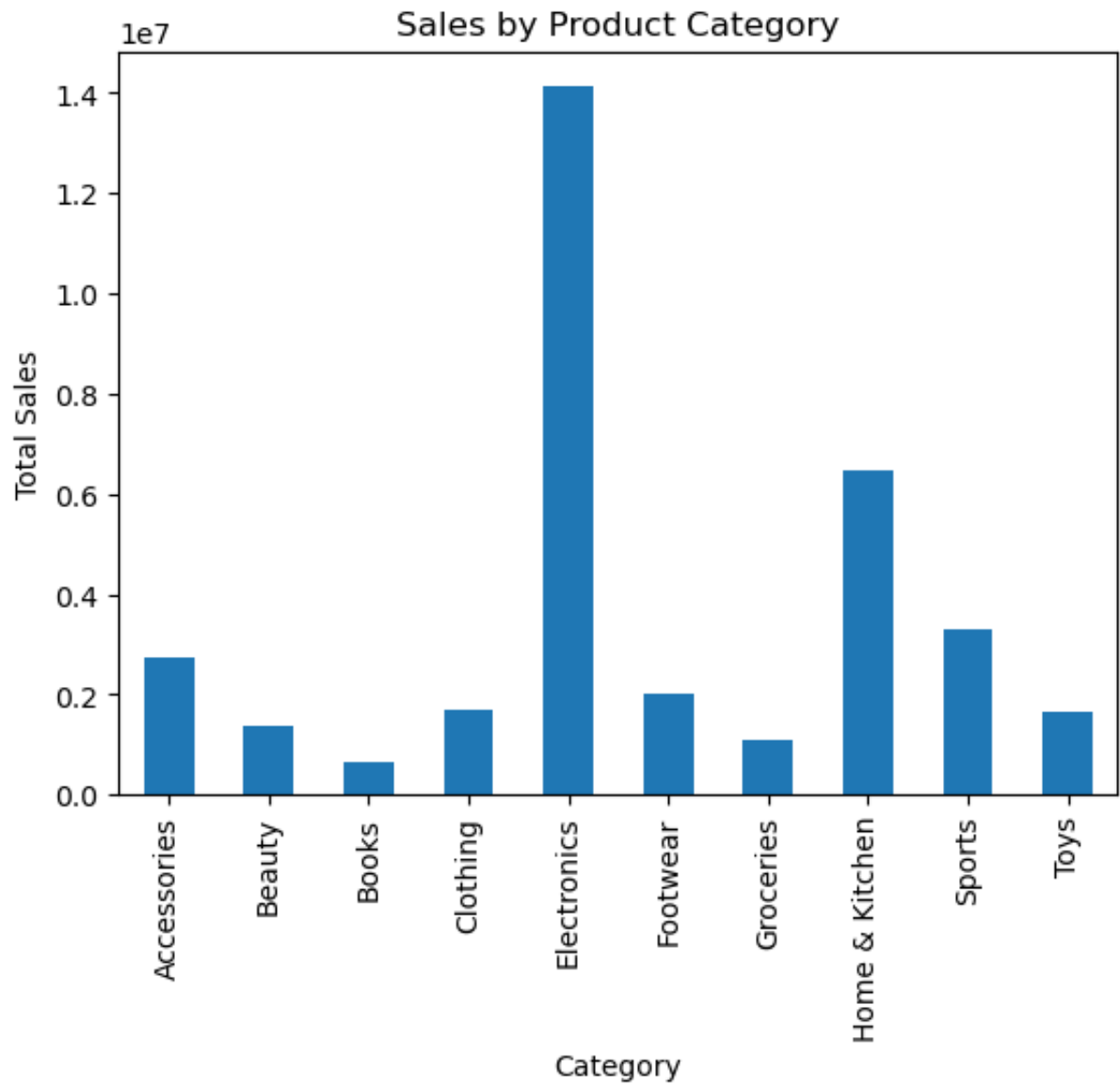
```
In [66]: sales_by_month = df.groupby(  
        df["purchase_date"].dt.month  
        )["purchase_amount"].sum()  
  
ax = sales_by_month.plot(kind="bar")  
  
plt.title("Sales by Month")  
plt.xlabel("Month")  
plt.ylabel("Total Sales")  
  
ax.ticklabel_format(style="plain", axis="y")  
  
plt.show()
```



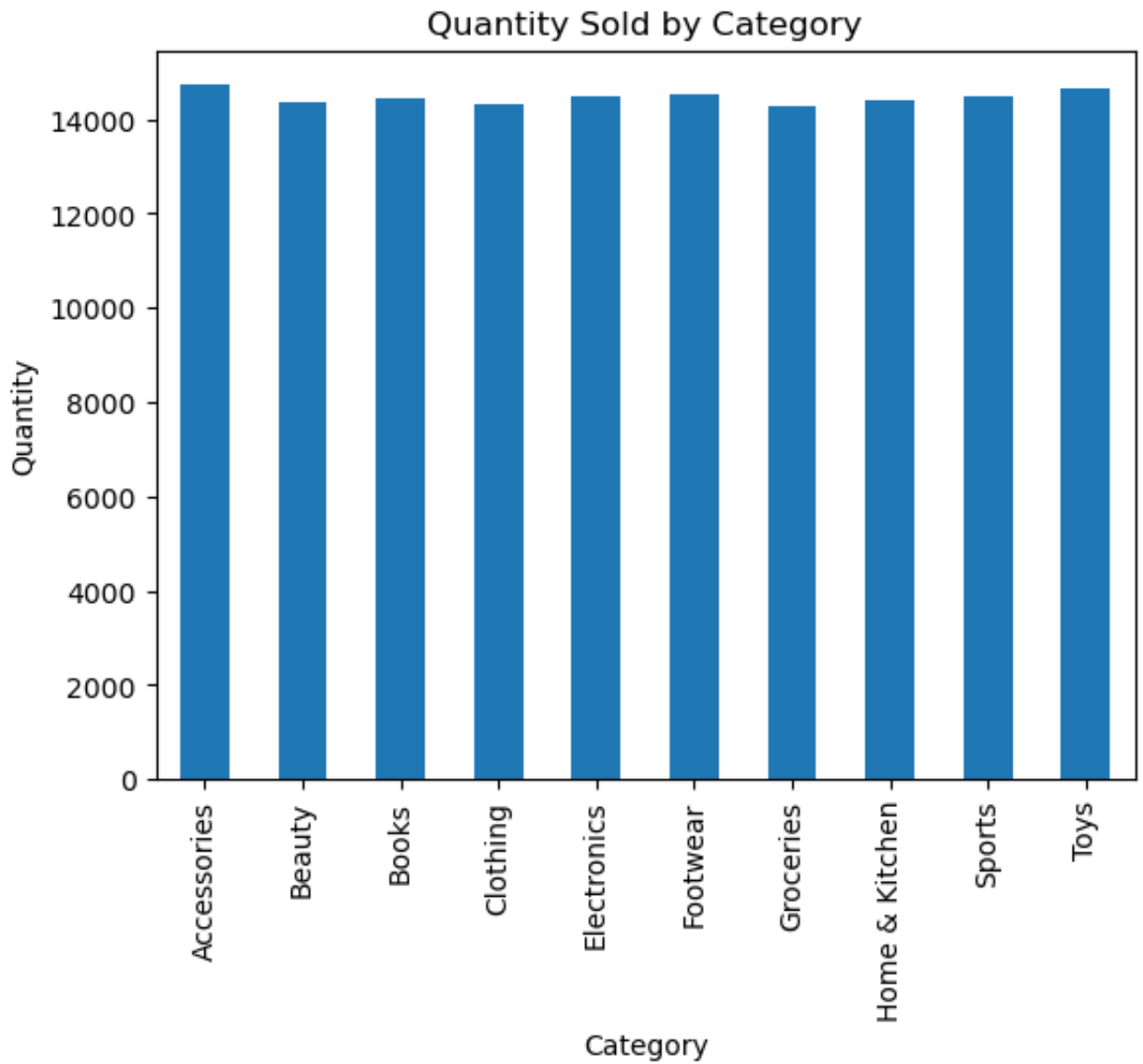
```
In [67]: quantity_by_category = df.groupby(  
        "product_category"  
        )["quantity"].sum()  
  
        quantity_by_category.plot(kind="bar")  
  
        plt.title("Quantity Sold by Category")  
        plt.xlabel("Product Category")  
        plt.ylabel("Quantity Sold")  
        plt.xticks(rotation=45)  
  
        plt.show()
```



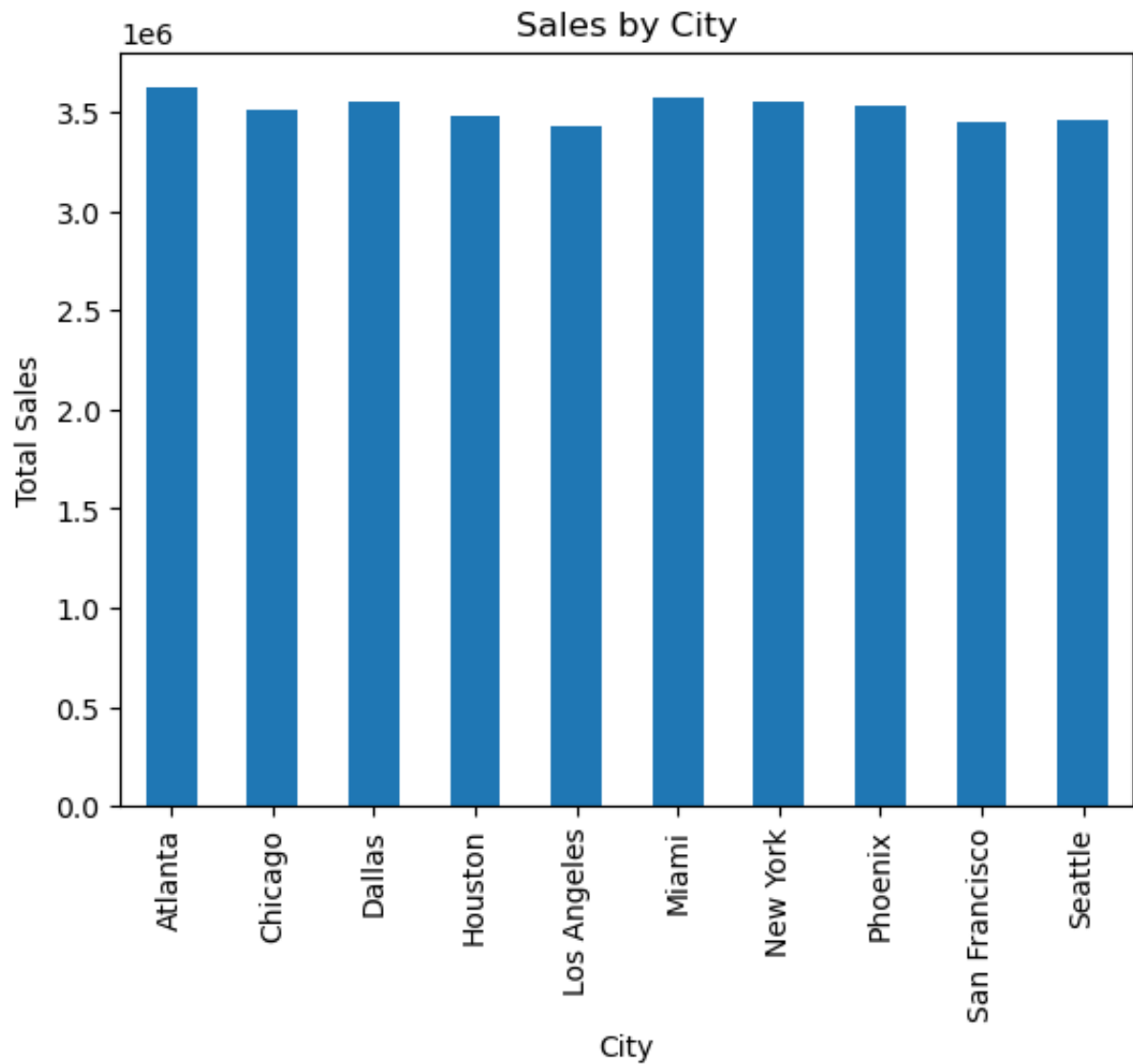
```
In [68]: sales_by_category = df.groupby(  
        "product_category"  
        )["purchase_amount"].sum()  
  
        sales_by_category.plot(kind="bar")  
  
        plt.title("Sales by Product Category")  
        plt.xlabel("Category")  
        plt.ylabel("Total Sales")  
  
        plt.show()
```



```
In [69]: quantity_by_category = df.groupby(  
        "product_category"  
    )["quantity"].sum()  
  
    quantity_by_category.plot(kind="bar")  
  
    plt.title("Quantity Sold by Category")  
    plt.xlabel("Category")  
    plt.ylabel("Quantity")  
  
    plt.show()
```



```
In [70]: sales_by_city = df.groupby(  
        "city"  
        )["purchase_amount"].sum()  
  
        sales_by_city.plot(kind="bar")  
  
        plt.title("Sales by City")  
        plt.xlabel("City")  
        plt.ylabel("Total Sales")  
  
        plt.show()
```

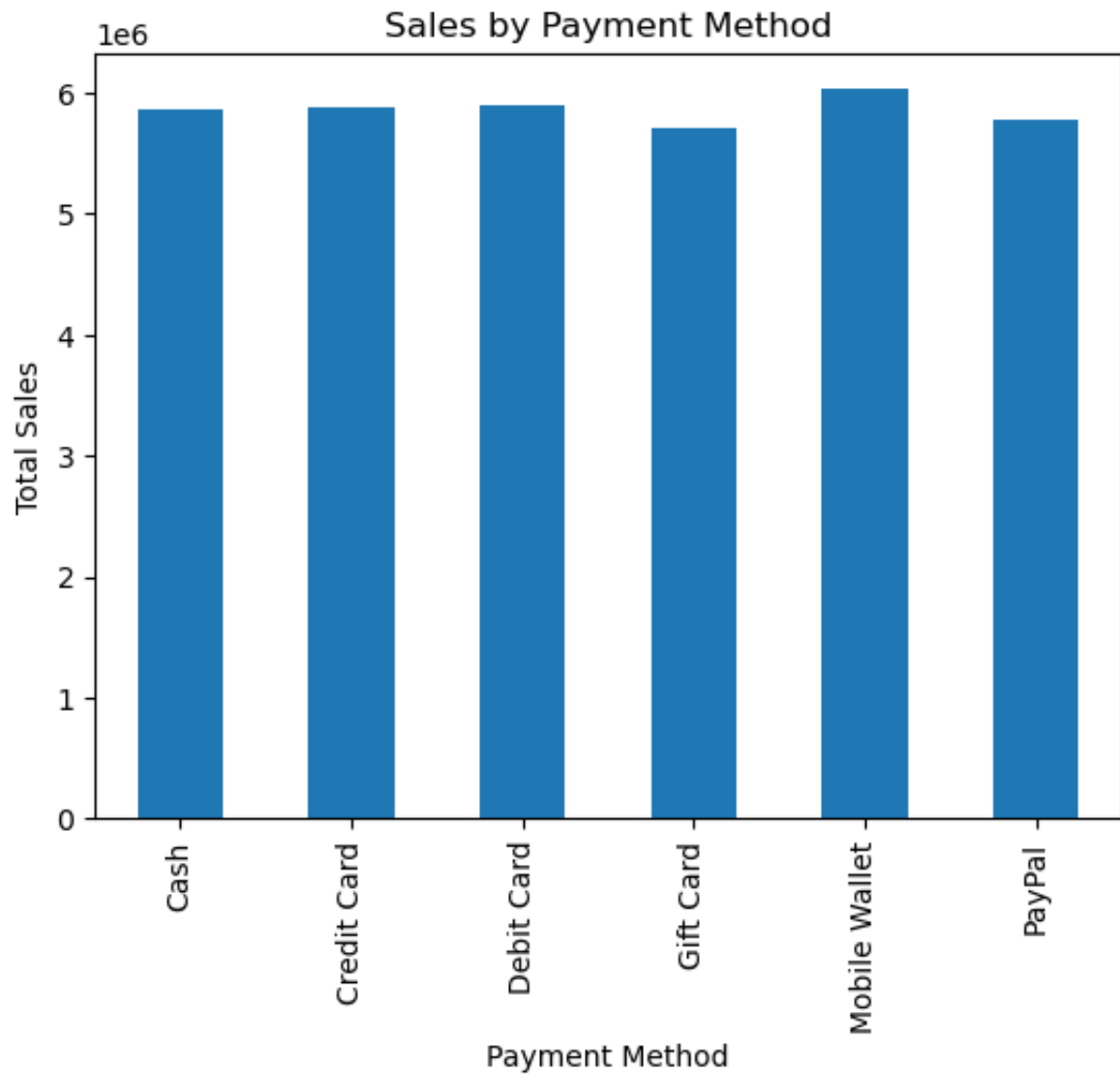



```
In [71]: sales_by_payment = df.groupby(
          "payment_method"
        )["purchase_amount"].sum()

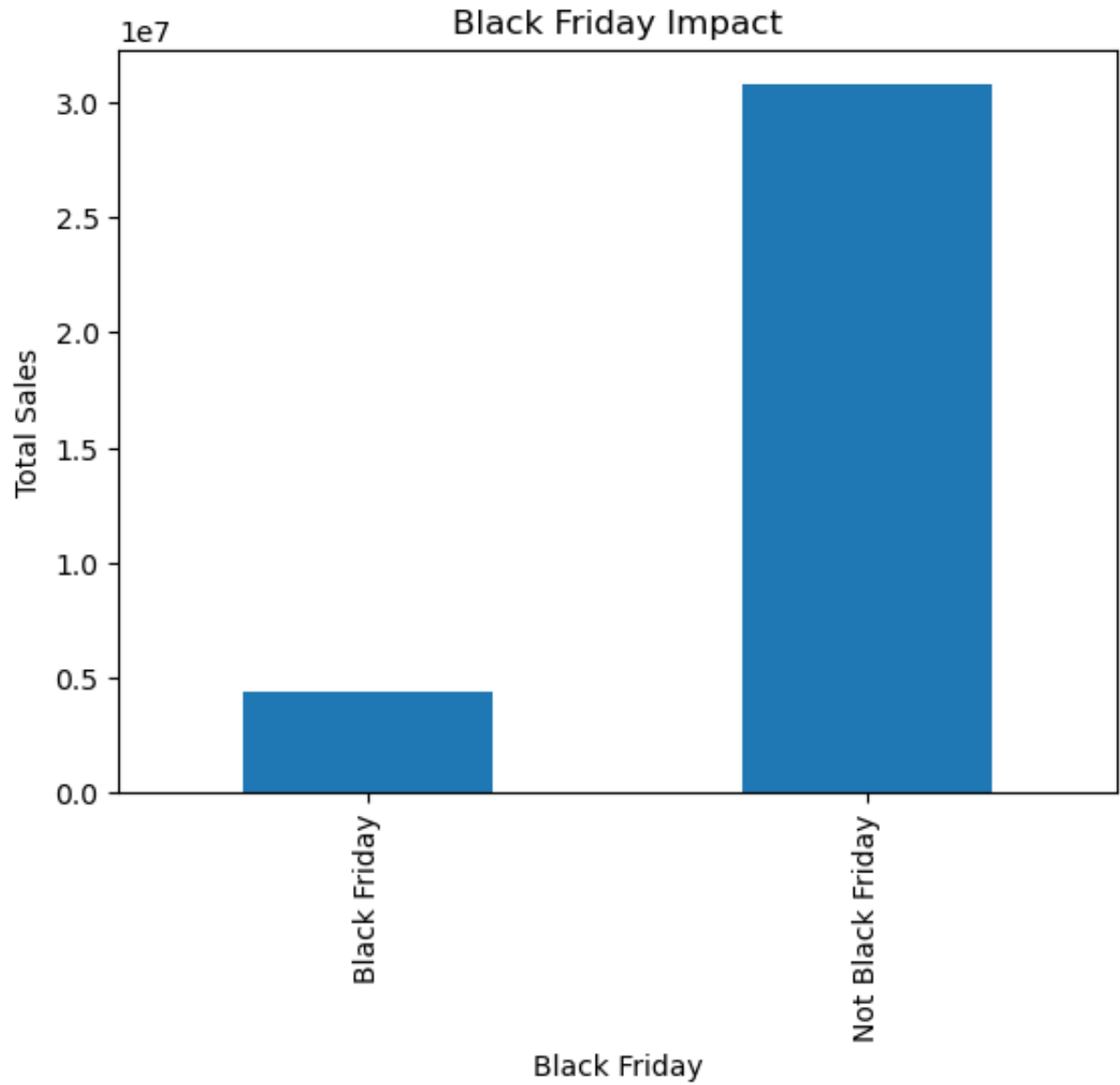
sales_by_payment.plot(kind="bar")

plt.title("Sales by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Total Sales")

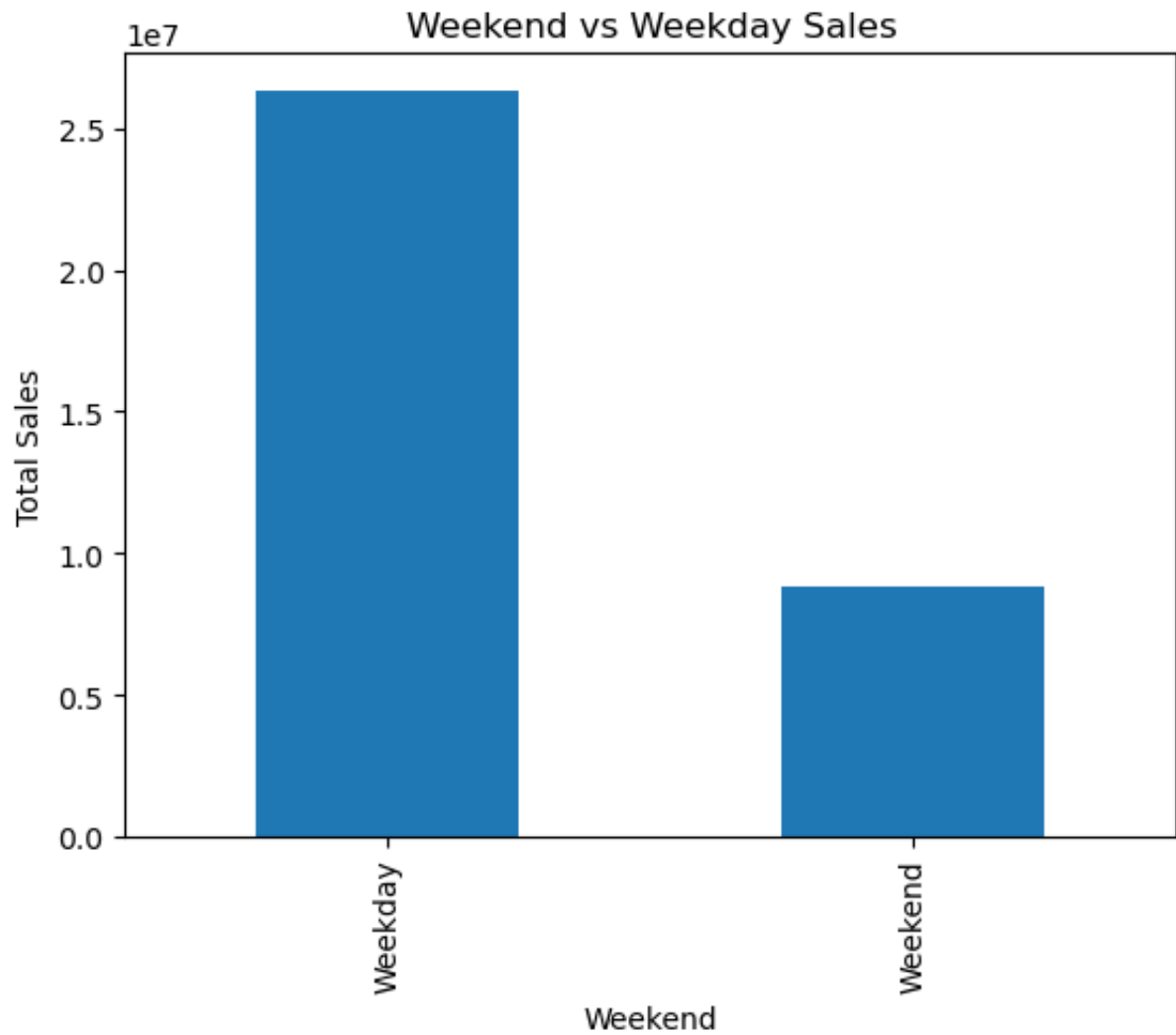
plt.show()
```



```
In [77]: black_friday_sales = df.groupby(  
        "is_black_friday"  
        )["purchase_amount"].sum()  
  
        black_friday_sales.plot(kind="bar")  
  
        plt.title("Black Friday Impact")  
        plt.xlabel("Black Friday")  
        plt.ylabel("Total Sales")  
  
        plt.show()
```



```
In [76]: weekend_sales = df.groupby(  
        "is_weekend"  
        )["purchase_amount"].sum()  
  
        weekend_sales.plot(kind="bar")  
  
        plt.title("Weekend vs Weekday Sales")  
        plt.xlabel("Weekend")  
        plt.ylabel("Total Sales")  
  
        plt.show()
```



```
In [74]: df["is_black_friday"] = df["is_black_friday"].replace({
          0: "Not Black Friday",
          1: "Black Friday"
        })
```

```
In [75]: df["is_weekend"] = df["is_weekend"].replace({
          0: "Weekday",
          1: "Weekend"
        })
```

```
In [55]: df.groupby("product_category")["purchase_amount"].mean().round(2)
```

```
Out[55]: product_category
Accessories      271.23
Beauty           137.70
Books             65.92
Clothing         170.36
Electronics     1416.03
Footwear         199.26
Groceries        108.41
Home & Kitchen   653.56
Sports           331.18
Toys             166.02
Name: purchase_amount, dtype: float64
```

```
In [58]: df.groupby("city")["purchase_amount"].mean().round(2)
```

```
Out[58]: city
Atlanta      359.14
Chicago      352.86
Dallas       355.15
Houston      346.66
Los Angeles  352.96
Miami        355.40
New York     350.96
Phoenix      351.48
San Francisco 345.64
Seattle      342.36
Name: purchase_amount, dtype: float64
```

```
In [60]: df.groupby("product_category")["discount_amount"].mean().round(2)
```

```
Out[60]: product_category
Accessories      67.63
Beauty           34.79
Books            16.69
Clothing         42.98
Electronics     355.89
Footwear         50.39
Groceries        26.98
Home & Kitchen   162.77
Sports           83.12
Toys            41.15
Name: discount_amount, dtype: float64
```

```
In [61]: df.groupby("city")["discount_amount"].mean().round(2)
```

```
Out[61]: city
Atlanta      90.35
Chicago      87.92
Dallas       87.50
Houston      84.35
Los Angeles  88.28
Miami        89.03
New York     88.58
Phoenix      90.21
San Francisco 86.34
Seattle      88.03
Name: discount_amount, dtype: float64
```

```
In [62]: charts = {
    "sales_by_category.png": df.groupby("product_category")["purchase_amo
    "sales_by_city.png": df.groupby("city")["purchase_amount"].sum(),
    "sales_by_payment.png": df.groupby("payment_method")["purchase_amount
    "quantity_by_category.png": df.groupby("product_category")["quantity"
}

for filename, data in charts.items():
    plt.figure(figsize=(8,4))
    data.plot(kind="bar")
    plt.tight_layout()
    plt.savefig(filename)
    plt.close()
```

```
In [ ]:
```